Complementarity in Social Measurement: A Partition-Logic Approach

Karl Svozil*

Institute for Theoretical Physics, TU Wien, Wiedner Hauptstrasse 8-10/136, A-1040 Vienna, Austria

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Abstract

Partition logics—non-Boolean operational event structures obtained by pasting Boolean algebras—provide a natural formal language for situations in which a system has a definite latent state but can be accessed only through mutually incompatible coarse-grained modes of observation. We show how this structure can be instantiated in six idealized but substantively interpretable social-science settings: personnel assessment, survey framing, clinical diagnosis, espionage coordination, legal pluralism, and organizational auditing. For each case we identify the latent state space, the observational contexts as partitions, and the shared atoms that intertwine contexts, yielding instances of the L_{12} bowtie, triangle, pentagon, and two-context partition logics. These examples make precise a notion of social complementarity: different modes of inquiry can be incompatible even though the underlying system remains fully value-definite. Complementarity in this sense does not entail Kochen–Specker contextuality or ontic indeterminacy. We further distinguish the classical probabilities induced by distributions over latent states from quantum-like Born probabilities available when the same exclusivity graph admits a suitable orthogonal representation, and from additional "exotic" weights supported by some odd cyclic logics. The framework thus separates operational logical structure from probabilistic realization and provides testable benchmarks for quantum-cognition models.

Keywords: partition logic; social complementarity; social measurement; quantum cognition; diagnostic contexts; non-Boolean probability

1 Introduction

Complementarity, the impossibility of simultaneously observing all relevant properties of a system, is usually associated with quantum mechanics. Niels Bohr

*karl.svozil@tuwien.ac.at

famously argued that the wave and particle aspects of a quantum object cannot be revealed in a single experimental arrangement. Yet the *formal* structure underlying complementarity—a collection of Boolean "classical mini-universes" pasted together at shared elements—also appears in idealized models of social-science measurement. In these situations the system under study (an applicant, a survey respondent, an organization) may possess fully determinate properties, but the observer is forced to choose among incompatible modes of inquiry, each of which reveals only a coarse-grained view of the true state.

The mathematical framework that captures this situation is the *partition logic* Dvurečenskij, Pulmannová, and Svozil (1995); Schaller and Svozil (1994); Svozil (1993, 2005). A partition logic starts from a finite set of underlying types $S_n = \{1, 2, \ldots, n\}$. Each "context" or "block" is a partition of S_n into groups: elements in the same group are those types that cannot be distinguished when that particular observational context is chosen. Two contexts are then "pasted" together whenever they share an identical group of types—an "intertwining atom." The resulting operational structure is, in general, non-Boolean: only the Boolean operations internal to a single context are operationally accessible. The full ambient set algebra 2^{S_n} still exists, but the social observer does not have access to all its distinctions in one measurement arrangement.

The purpose of this paper is to make this abstract formalism concrete through six stylized social-science scenarios spanning organizational psychology, political science, clinical psychology, security studies, comparative law, and public administration. For each scenario we will answer three questions explicitly:

1. What are the **types** (elements of S_n), and what do they represent in the social world?
2. What are the **contexts** (partitions), and what observational procedure does each one correspond to?
3. What are the **intertwining atoms**—that is, which observed categories occur identically in more than one context—and what substantive feature of the social situation explains this cross-context identification?

A central methodological point of the paper is that three levels must be kept distinct. First, there is the latent Boolean space 2^S, in which all subsets of the finite type space exist as ordinary set-theoretic events. Second, there is the restricted operational event structure obtained by retaining only those coarse-grained events that correspond to actually available measurement contexts; this pasted structure may be non-Boolean even though it is represented by subsets of S. Third, there are probability models on the operational structure. The classical model comes from probability distributions over S. Other models, such as Born-rule models associated with orthogonal representations or "exotic" odd-cycle weights, may live on the same abstract exclusivity structure but need not arise from any distribution over the intended latent social profiles.

This distinction is important for the quantum-cognition program Busemeyer and Bruza (2012); Pothos and Busemeyer (2013). The same exclusivity graph can support different probability models. Classical partition models yield convex combinations of point-evaluation states. Quantum systems can yield Born probabilities. Some odd cyclic logics support additional fractional weights that lie outside the classical convex hull. Thus the operational graph alone does not determine the probability theory; the resource that realizes the graph does.

Scope and claims of novelty. The mathematical objects used below—partition logics, the L_{12} bowtie, Wright's triangle and pentagon logics, the KCBS bound, and Wright's exotic odd-cycle weights—are all pre-existing. What this paper aims to contribute is (i) an explicit three-level taxonomy separating the latent Boolean ontology, the restricted operational event structure, and the probabilistic model; (ii) six concrete social-science instantiations selected to span the qualitatively different structures available at low complexity (two-context bowtie, three-context cycle, five-context cycle, classical common-cause correlations, and maximally complementary two-context inspection); and (iii) explicit empirical templates—most notably a pentagon-structured survey design with a sharp classical-versus-Born prediction—that translate the formalism into preregistered tests of quantum-cognitive hypotheses. The framework deliberately does *not* commit to any particular ontology of cognition; it provides a language in which competing ontologies make distinct, falsifiable predictions.

Three notions of complementarity. The word "complementarity" will be used below in three related but formally distinct senses, and it is useful to separate them at the outset:

(C1) *Logical/operational complementarity.* The pasted event structure is non-Boolean: cross-context set-theoretic intersections exist in the ambient algebra 2^S but are not operationally licensed within any single context.

(C2) *Dynamical or disturbance-based complementarity.* Probing the system with one instrument changes its state in a way that is relevant to subsequent probes, so that joint or sequential administration of contexts on a single system is not equivalent to querying a fixed underlying state.

(C3) *Framing-based complementarity.* Distinct contexts conceptually carve the same latent space along different dimensions, so that even an idealized non-disturbing observer would obtain genuinely different categorical descriptions.

These three notions can come apart: (C2) and (C3) are conceptually independent of each other, and either can hold without (C1) when the contexts share no intertwining atoms. Most of our scenarios combine all three; the auditing example of Section 8 foregrounds (C2) while exhibiting only a trivial form of (C1).

While the history and sociology of the quantum mechanics community are fascinating subjects in their own right Cabello (2017); Howard (2004), an analysis of these aspects falls outside the scope of this paper.

Section 2 reviews the formalism. Sections 3–8 present the six scenarios. Section 9 discusses probability models and their interpretation. Section 10 concludes.

2 Preliminaries: Partition logics and their probabilities

Before delving into the specific social-science scenarios, we first establish the formal mathematical framework of partition logics, their graphical representations, and the probability models they may support. This provides a rigorous foundation for modeling complementarity in social measurements.

2.1 The elements: latent states, observed categories, and pasting

A partition logic starts from a finite set $S = \{s_1, s_2, \ldots, s_n\}$ of *latent states*. Each latent state is one fully specified possibility in the model: for example, a voter who is economically left-leaning and culturally conservative, or a household that is above the poverty line but materially deprived, or a job applicant whose true competence profile is "strong analytical skills, weak interpersonal skills." We assume that, in reality, the system under study is exactly one latent state $s \in S$, even if the observer cannot always determine which one. The substantive interpretation of a latent state—what it means for a person, household, student, or organization to be in state s—is called the *latent social profile* associated with s. The word "latent" signals only that the full profile is not directly read off from any single observational procedure; it does not imply that the profile is indeterminate or unreal.

The set S of latent states captures the ontology of the situation: what the system truly is within the model. By contrast, what can be inferred about the system—and what distinctions cannot be drawn with a particular instrument—belongs to the epistemology. That epistemological structure is encoded by partitions of S.

A *context* is a partition $\mathcal{C} = \{B_1, B_2, \ldots, B_m\}$ of S into m nonempty, pairwise disjoint subsets whose union is S. Each context represents one available mode of observation: a survey item, an administrative classification rule, a diagnostic instrument, a coding scheme, or a framing device. Each cell $B_j \subseteq S$ of the partition is called an *observed category* or *atom* of that context. An observed category is the coarsest unit of information that the chosen context can deliver: when the observer selects context $\mathcal{C}$, all latent states within the same observed category B_j produce indistinguishable outcomes. The observer learns only which observed category contains the true latent state, not which specific latent state it is. Thus the partition encodes the resolving power—and the limitations—of a particular mode of inquiry.

To summarize the ontology–epistemology mapping:

- A **latent state** $s \in S$ is one exact underlying possibility.
- A **latent social profile** is the substantive social-science interpretation of that latent state.
- A **context** $\mathcal{C}$ is a partition of S induced by a particular instrument, survey frame, legal code, or audit directive.
- An **observed category** $B_j \subseteq S$ is one cell of that partition—the group of latent states that the chosen context cannot tell apart.

Two contexts $\mathcal{C}$ and $\mathcal{C}'$ may share an observed category: a subset $B \subseteq S$ may appear as an atom in both partitions. Such a shared observed category is called an *intertwining atom*. It is the formal point at which two Boolean subalgebras—each representing the propositional logic of one context, in which all distinctions are classical and simultaneously decidable—are pasted together to form the restricted operational partition logic Gerelle, Greechie, and Miller (1974); Svozil (2020); Wright (1978).

The resulting operational structure is non-Boolean in the following precise sense. Although all observed categories are subsets of the global latent space S, only joins and complements within a common context are operationally licensed. Cross-context set-theoretic intersections exist in the ambient Boolean algebra 2^S, but they need not correspond to any jointly performable observation. Thus the non-Booleanity is not an ontological failure of 2^S; it is an operational feature of the restricted pasted event structure. In the terminology introduced in Section 1, this is logical/operational complementarity (C1); whether the same scenario also exhibits disturbance-based (C2) or framing-based (C3) complementarity is a separate substantive question for each instantiation.

Complementarity manifests itself in the fact that different contexts partition the latent-state space differently: observed categories that are resolved by one context may be conflated by another. Crucially, every latent state retains its determinate identity throughout. The complementarity is a limitation on the observer's ability to ascertain the latent social profile, not an indeterminacy in the profile itself.

2.2 Graphical representations

Two equivalent graphical notations are common. In a hypergraph (often also referred to as a Greechie orthogonality diagram) Bretto (2013); Greechie (1971), each context is drawn as a smooth line or hyperedge and each atom as a vertex on that line; intertwining atoms sit at the intersection of two or more lines. In the exclusivity graph, vertices are atoms and edges connect atoms that are mutually exclusive, i.e., that belong to the same context.

Uniform hypergraphs are characterized by an identical number of vertices on every hyperedge. In standard quantum logic, an n-dimensional Hilbert space naturally generates an n-uniform hypergraph, because every orthonormal basis contains n elements. In the social sciences, however, this strict uniformity is not required. Complementary observational procedures may yield maximal sets of mutually exclusive outcomes of varying sizes; formally, the blocks comprising the partition logic may be Boolean algebras of different orders Greechie (1971). This structural departure from standard quantum mechanics is not, by itself, an obstacle to the partition-logic framework. The admissibility conditions for probabilities are imposed block by block. Because the concrete scenarios presented in this paper rely on simple uniform hypergraphs, a fuller discussion of non-uniform structures is deferred to future work.

2.3 Probability models on the same operational structure

Henceforth, any admissible probability weight on the operational logic must satisfy two Kolmogorov-type conditions on each block (classical mini-universe) Gleason (1957):

1. **Additivity/exclusivity**: the probability of a union of mutually exclusive events within the same block is the sum of their individual probabilities; and
2. **Normalization/completeness**: the probabilities of all mutually exclusive atoms comprising a block sum to 1.

A *two-valued weight* or *dispersion-free state* on the operational logic is a function $v\colon \text{atoms} \to \{0,1\}$ that assigns value 1 to exactly one atom per context. When the partition logic is represented concretely by subsets of a latent space S, each $s \in S$ induces a point-evaluation two-valued state:

$$v_s(B) = \begin{cases} 1, & s \in B, \\ 0, & s \notin B. \end{cases}$$

Conversely, an abstract pasted logic may possess additional two-valued weights that do not correspond to any element of the chosen latent space. In the social-science interpretation below we treat the point-evaluation states as the intended ontic states, whereas the larger set of abstract two-valued weights belongs to the purely logical representation.

Classical probabilities. Given a population of systems (applicants, respondents, patients, organizations), the fraction of each latent type defines a probability distribution over S. The resulting probabilities on observed categories are

$$P(B_j) = \sum_{s_i \in B_j} \lambda_i, \qquad \lambda_i \geq 0, \qquad \sum_i \lambda_i = 1.$$

These are convex combinations of point-evaluation states. In the language of exclusivity graphs, the corresponding noncontextual classical set is a stable-set or vertex-packing type polytope, depending on the precise graph convention Grötschel, Lovász, and Schrijver (1986).

Quantum-like Born-rule probabilities. If the exclusivity graph G admits a suitable faithful orthogonal representation (FOR)—an assignment of vectors to atoms such that exclusive atoms are represented by orthogonal vectors and nonexclusive atoms by nonorthogonal ones Cabello, Severini, and Winter (2014); Lovász (1979)—one can define a Born-type probability by choosing a unit state vector $|c\rangle$ and setting

$$P_{\mathrm{Born}}(c, v_i) = |\langle c|v_i\rangle|^2. \tag{1}$$

For the probabilities to normalize on each context, it is not enough that exclusive atoms be pairwise orthogonal; the vectors or projectors associated with each context must also form a complete orthonormal resolution of the identity on the relevant subspace. Under this additional condition, the Pythagorean theorem guarantees that probabilities within each block add to one.

With the standard exclusivity-graph convention, the corresponding quantum set is captured by a Lovasz-theta-type relaxation $\mathrm{TH}(G)$, up to the common convention of whether G or its complement is used Grötschel et al. (1986). In general this quantum-like set can strictly contain the classical polytope: there exist Born-rule distributions that cannot be realized by any classical mixture of latent point states.

Exotic weights. Some abstract pasted logics also support admissible probability weights that are neither classical mixtures of point states nor necessarily Born-rule weights. Odd cyclic logics, such as the triangle and pentagon/pentagram, support a well-known fractional weight assigning 1/2 to each cyclic intertwining atom Gerelle et al. (1974); Wright (1978). We discuss this explicitly in Section 5.5.

The crucial point is that the same formal graph can be implemented by different resources. A social-science black box with determinate latent types yields classical probabilities from distributions over S. A quantum resource can yield Born probabilities. An abstract pasted logic may also admit additional admissible weights. The graph alone does not determine the probability theory; the resource inside the box does Svozil (2020).

Context-wise softmax probabilities. We close with a construction that is often used as a response model in the social and cognitive sciences, but that must be handled carefully in partition-logical applications. Unlike a probability weight on a pasted logic, a softmax rule is naturally defined context by context. It automatically normalizes probabilities within each context, but it does not by itself enforce the cross-context identifications imposed by intertwining atoms.

Suppose that, for each context $\mathcal{C}$ and each atom $B_j \in \mathcal{C}$, the observer or respondent assigns a real-valued score

$$u_{\mathcal{C}}(B_j) \in \mathbb{R}.$$

This score may represent utility, salience, diagnostic strength, response tendency, evidential support, or any other context-dependent propensity associated with the observed category B_j. A standard normalization is the softmax rule

$$P_{\beta,\mathcal{C}}(B_j) = \frac{\exp\big(\beta u_{\mathcal{C}}(B_j)\big)}{\sum_{B_k \in \mathcal{C}} \exp\big(\beta u_{\mathcal{C}}(B_k)\big)}, \tag{2}$$

where $\beta \geq 0$ is an inverse-temperature or discrimination parameter. For $\beta = 0$, all atoms in the context are assigned equal probability. As $\beta \to \infty$, the distribution concentrates on the atom or atoms with maximal score.

The softmax rule automatically satisfies normalization within each context,

$$\sum_{B_j \in \mathcal{C}} P_{\beta,\mathcal{C}}(B_j) = 1,$$

and it respects additivity for unions of atoms inside the same Boolean block, since the probability of a coarse-grained event $E = \bigcup_{j \in J} B_j$ is obtained by summing the corresponding atom probabilities:

$$P_{\beta,\mathcal{C}}(E) = \sum_{j \in J} P_{\beta,\mathcal{C}}(B_j).$$

In this sense softmax yields, for each context separately, an ordinary normalized distribution over that context's observed categories.

It is important, however, that this is weaker than being a probability weight on the pasted operational logic. A weight on the pasted logic is a *single* function on atoms. Hence an intertwining atom—one subset $B \subseteq S$ shared by two or more contexts—must receive *one and the same* probability value wherever it occurs. A context-dependent score family $u_{\mathcal{C}}$ generically assigns different probabilities to the same intertwining atom in its different contexts, because both the scores and the normalizing denominators may depend on $\mathcal{C}$. Free-score softmax is therefore not, in general, a probability weight on the partition logic; it is a family of separately normalized per-context distributions. Promoting it to a genuine weight requires an additional *marginal-consistency* condition: whenever B is an atom of both $\mathcal{C}$ and $\mathcal{C}'$, one must have

$$P_{\beta,\mathcal{C}}(B) = P_{\beta,\mathcal{C}'}(B).$$

Only under such a tying condition does the construction define a probability weight on the operational pasted structure.

This also shows that softmax should not be read as an intermediate probability model lying between the classical and Born-rule sets. Without a tying condition, Eq. (2) imposes no cross-context constraint: each context may be fitted independently. On a cyclic structure such as the pentagon, for example, the probabilities assigned to the five intertwining atoms can be made large in their respective contexts, so their sum may exceed not only the classical bound (6) but also the quantum-like bound (7). Such an excess would not, by itself, indicate a stronger-than-quantum resource; it would simply show that the per-context response model has not implemented the pasted exclusivity structure.

The construction reduces to an ordinary classical partition model only under stronger conditions. If there exists a single latent distribution

$$\lambda_i \geq 0, \qquad \sum_i \lambda_i = 1,$$

such that the resulting atom probabilities satisfy

$$P_{\beta,\mathcal{C}}(B_j) = \sum_{s_i \in B_j} \lambda_i$$

for every atom B_j in every context $\mathcal{C}$, then the softmax probabilities are merely a reparametrization of classical latent-state marginals. Equivalently, for $\beta > 0$, one may choose scores of the form

$$u_{\mathcal{C}}(B_j) = \frac{1}{\beta} \log \left(\sum_{s_i \in B_j} \lambda_i \right) + a_{\mathcal{C}},$$

where $a_{\mathcal{C}}$ is an arbitrary context-dependent additive constant, with the usual limiting conventions when some probabilities vanish. In that case the softmax normalization adds no new probabilistic structure.

The substantive content of softmax as a modelling tool lies in the gap between these two extremes. In survey research, $u_{\mathcal{C}}(B_j)$ might encode the salience of a response category under a particular frame; in clinical diagnosis, the evidential strength of a diagnostic category under a particular instrument; in personnel assessment, the relative attractiveness of different classifications under a given screening procedure. The softmax map turns such scores into well-defined within-context probabilities, but whether those probabilities can be glued together into one global distribution over the intended latent profiles is a further question that softmax normalization alone does not answer. In particular, context-wise softmax normalization cannot replace the classical latent-state inequalities such as (6): those inequalities are precisely the test of whether locally normalized response probabilities admit a common latent-state explanation.

3 Scenario 1: Complementary personnel assessments (L_{12})

In what follows, we develop several concrete enactments of the partition-logical formalism introduced above. We emphasize at the outset that these are deliberately stylized models. Each scenario isolates a single structural feature—a particular partition logic together with an idealized motivation for forced choice between contexts—in order to make the formalism concrete. We do not claim that any of the scenarios captures the full complexity of real personnel selection, real surveys, or real clinical practice. The question we ask is rather: *if* a measurement situation has the indicated structure, what does partition-logic analysis predict? Such idealizations are common in formal modelling and serve to sharpen the comparison between classical and nonclassical probability models.

For related discussions and additional examples, we refer the reader to the extensive works of Aerts and colleagues Aerts (2024); Aerts and Aerts (1995); Aerts, Aerts, and Gabora (2009); Aerts and Aerts Arguëlles (2022); Aerts, Aerts Arguëlles, Beltran, and Sozzo (2023); Aerts, Aerts Arguëlles, and Sozzo (2025); Aerts and Gabora (2005); Aerts, Gabora, and Sozzo (2013), Khrennikov and colleagues Haven and Khrennikov (2013); A. Khrennikov (2009, 2015, 2020a, 2020b); A. Khrennikov, Basieva, Pothos, and Yamato (2018); A. Y. Khrennikov (2010), and numerous scholars Aerts, D'Hooghe, and Haven (2010); Arioli and Valente (2021); Asano, Khrennikov, Ohya, Tanaka, and Yamato (2015); Ashtiani and Azgomi (2015); Aspalter (2024a, 2024b, 2025); Atmanspacher and Römer (2012); Franco (2009); Holtfort and Horsch (2023); Lambert-Mogiliansky and Busemeyer (2012); Lambert-Mogiliansky, Zamir, and Zwirn (2009); Meghdadi, Akbarzadeh-T., and Javidan (2022); Orrell (2016, 2020); Wendt (2015); Yukalov and Sornette (2010, 2014, 2018). These authors have explored quantum-like structures across cognition, decision-making, social systems, and game theory Eisert, Wilkens, and Lewenstein (1999); Sanz-Martín, Rivas, Clavijo-Buriticá, Herrera, and Parra-Domínguez (2025).

3.1 What are the types?

A firm's human resources department faces a pool of job applicants. Each applicant has a true latent competence profile, falling into one of four categories:

- **Type 1 ("Star"):** Excellent both in analytical reasoning and in interpersonal communication.
- **Type 2 ("Analyst"):** Strong analytical skills but weaker interpersonal skills.
- **Type 3 ("Communicator"):** Strong interpersonal skills but weaker analytical ability.
- **Type 4 ("Developing"):** Still developing in both dimensions.

Thus $S_4 = \{1, 2, 3, 4\}$, and every applicant is exactly one type. The uncertainty is the HR department's, not the applicant's.

3.2 What are the contexts?

Two assessment instruments are available:

- **Written aptitude test (C_W):** A timed analytical exam. It can identify Stars (type 1) by their top scores and Analysts (type 2) by their strong-but-not-top scores, but it cannot distinguish Communicators from

Developing applicants, because both score poorly on a purely analytical task. The three observable outcomes are therefore

$$\{1\}, \qquad \{2\}, \qquad \{3, 4\}.$$

- **Behavioral interview (C_I):** A structured interpersonal exercise. It can identify Stars (type 1) by their social poise and Communicators (type 3) by their good-but-not-top social skills, but it cannot distinguish Analysts from Developing applicants, because both perform weakly in interpersonal tasks. The three observable outcomes are

$$\{1\}, \qquad \{3\}, \qquad \{2, 4\}.$$

Crucially, suppose the two instruments cannot be administered independently to the same applicant within a single assessment window. Several stylized rationales support this restriction: a high-volume initial screening in which only one instrument fits the time budget per applicant; a research design in which applicants are randomized between conditions so that the joint test–interview sequence is never observed on the same person; or, more substantively, well-documented carryover effects in which one modality contaminates subsequent performance in the other—administering the written test first can induce test anxiety or stereotype threat that biases interview performance, while social priming from the interview can anchor subsequent test-taking Tversky and Kahneman (1974). We do not claim that actual HR practice is forced into such a single-instrument regime—multimethod assessment is in fact common—but the stylized restriction is what isolates the L_{12} structure and gives the framework its predictive content.

Formally, the two contexts are represented by the partitions

$$C_W = \{\{1\}, \{2\}, \{3, 4\}\}, \qquad C_I = \{\{1\}, \{3\}, \{2, 4\}\}. \tag{3}$$

In epistemic terms, these are the three-outcome classifications

$$\{\text{Outstanding}, \text{Analytical-only}, \text{Low-test-score}\}$$

and

$$\{\text{Outstanding}, \text{Social-only}, \text{Low-interview-score}\}.$$

Table 1: Response profiles for four applicant types under two complementary instruments. Each row is an applicant type; each column shows how that type appears under the given instrument.

Applicant type	Written test outcome	Interview outcome
1: “Star”	Outstanding	Outstanding
2: “Analyst”	Analytical-only	Low-interview-score
3: “Communicator”	Low-test-score	Social-only
4: “Developing”	Low-test-score	Low-interview-score

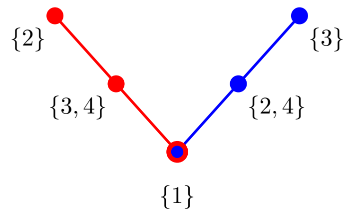

Figure 1: Greechie diagram of the L_{12} personnel-assessment logic. Each line represents one context. The written-test context (left) carries the observed categories $\{2\}$, $\{3,4\}$, and $\{1\}$. The interview context (right) carries the observed categories $\{3\}$, $\{2,4\}$, and $\{1\}$. The observed category $\{1\}$ is the intertwining atom: it appears in both contexts because the Star profile is identifiable regardless of which instrument is chosen. The designation L_{12} counts the twelve elements of the pasted operational logic: each three-atom Boolean block contributes $2^3 = 8$ elements, and the two blocks share exactly 4 elements (the intertwining atom $\{1\}$, its block complement $\{2,3,4\} = \{3,4\} \cup \{2\} = \{2,4\} \cup \{3\}$ inside each block, together with $\emptyset$ and the unit), giving $8 + 8 - 4 = 12$. The label L_{12} does not refer to the full Boolean algebra 2^{S_4}, which has 16 elements.

3.3 What is the intertwining atom?

The atom $\{1\}$ appears identically in both partitions (Table 1). This is because the Star type excels on both dimensions, so regardless of which instrument is chosen, Stars are singled out. In the language of partition logic, $\{1\}$ is the intertwining atom at which the two three-element Boolean algebras are pasted together.

Note what is not shared: the low-performance label has a different meaning in each context—$\{3,4\}$ under the written test versus $\{2,4\}$ under the interview. Although the label is similar, the underlying sets of types are different, so these are not intertwining atoms. Only atoms corresponding to exactly the same set of latent states are identified.

3.4 What does the partition logic tell us?

The partition logic L_{12} (Fig. 1) encodes a precise tradeoff. By choosing the written test, the HR department can distinguish type 2 from types 3 and 4, at the cost of being unable to distinguish type 3 from type 4. By choosing the interview, it can distinguish type 3 from types 2 and 4, but now type 2 and type 4 are conflated. No single instrument resolves all four types. This is complementarity: information about one aspect of the applicant comes at the expense of information about the other. Yet every applicant has a determinate type; the partition logic captures an epistemic limitation of the observer, not an ontic indeterminacy of the system.

4 Scenario 2: Framing effects in public opinion surveys (Pentagon logic)

Our second scenario turns to political psychology. It shows how a carefully idealized survey design with strong framing and order effects can instantiate the cyclic structure of a pentagon partition logic.

4.1 What are the types?

A political psychologist studies a population whose members differ in their latent configuration of attitudes across five policy domains. Rather than continuous attitude scales, suppose for simplicity that the relevant attitudinal differences can be captured by $n = 11$ discrete respondent profiles, denoted $S_{11} = \{1, 2, \ldots, 11\}$. The choice of $n = 11$ is not arbitrary: it is the minimum cardinality of a latent space that supports the pentagon partition logic with all five contexts having three atoms each and all five intertwining atoms being distinct subsets of S (Wright, 1978, p. 267, Fig. 2). Smaller latent spaces force identifications that collapse the cyclic structure. Each profile specifies a particular pattern of beliefs about immigration, healthcare, taxation, education, and defense spending. For example:

- Profile 1 might be a consistent moderate who holds centrist views on all five issues.
- Profile 7 might be a libertarian who favors minimal government across all domains.
- Profile 10 might be a hawk-dove who supports high defense spending as well as generous social programs.

The eleven profiles are assumed exhaustive and mutually exclusive: every respondent is exactly one profile.

4.2 What are the contexts?

The five policy issues define five survey contexts. In each context, the respondent is asked a block of questions about one issue and the response is classified into one of three opinion clusters, such as supportive, moderate, and opposed. To idealize strong framing and order effects Moore (2002), suppose that the research design treats each issue frame as a separate experimental condition. Each respondent is assigned to one context only (between-subjects design), so that cross-context joint responses are not taken as observations of a single unchanged cognitive state.

The five contexts, adapted from the partition logic of the pentagon Gerelle et al. (1974); Svozil (2020); Wright (1978), are:

$$\begin{aligned} C_1 \text{ (immigration)} &= \big\{\{1,2,3\},\ \{4,5,7,9,11\},\ \{6,8,10\}\big\}, \\ C_2 \text{ (healthcare)} &= \big\{\{6,8,10\},\ \{1,2,4,7,11\},\ \{3,5,9\}\big\}, \\ C_3 \text{ (taxation)} &= \big\{\{3,5,9\},\ \{1,4,6,10,11\},\ \{2,7,8\}\big\}, \\ C_4 \text{ (education)} &= \big\{\{2,7,8\},\ \{1,3,9,10,11\},\ \{4,5,6\}\big\}, \\ C_5 \text{ (defense)} &= \big\{\{4,5,6\},\ \{7,8,9,10,11\},\ \{1,2,3\}\big\}. \end{aligned} \tag{4}$$

Each context partitions all eleven profiles into three clusters.

4.3 What are the intertwining atoms?

Consider contexts C_1 and C_2. The atom $\{6,8,10\}$ appears as one cluster in both contexts. Substantively, this means that profiles 6, 8, and 10 happen to look the same under the immigration frame and under the healthcare frame. The cluster $\{6,8,10\}$ is an intertwining atom precisely because the same subset of latent profiles is produced by two different observational contexts.

The five intertwining atoms, one per adjacent pair of contexts, form the outer vertices of the pentagon diagram (Fig. 2):

$$\{1,2,3\}, \quad \{6,8,10\}, \quad \{3,5,9\}, \quad \{2,7,8\}, \quad \{4,5,6\}.$$

Formally, the five cyclically intertwined contexts form a pentagon/pentagram logic (Wright, 1978, p. 267, Fig. 2) supporting eleven intended point-evaluation states. Constructing the five contexts from the occurrences of value 1 in the corresponding point states yields the partition logic

$$\begin{aligned} &\{\{\{1,2,3\},\{4,5,7,9,11\},\{6,8,10\}\}, \\ &\{\{6,8,10\},\{1,2,4,7,11\},\{3,5,9\}\}, \\ &\{\{3,5,9\},\{1,4,6,10,11\},\{2,7,8\}\}, \\ &\{\{2,7,8\},\{1,3,9,10,11\},\{4,5,6\}\}, \\ &\{\{4,5,6\},\{7,8,9,10,11\},\{1,2,3\}\}\}. \end{aligned} \tag{5}$$

4.4 Why the pentagon structure matters: classical vs. quantum-like bounds

Each intertwining atom is a subset of the eleven profiles. The classical probability that a randomly drawn respondent falls in any one of these subsets is simply the fraction of the population belonging to the corresponding profiles. The key constraint is that the five intertwining-atom subsets overlap in a restricted way: each latent profile belongs to at most two of the five intertwining atoms. Therefore, for every point state,

$$\sum_{j=1}^{5} \chi_{A_j}(s) \leq 2,$$

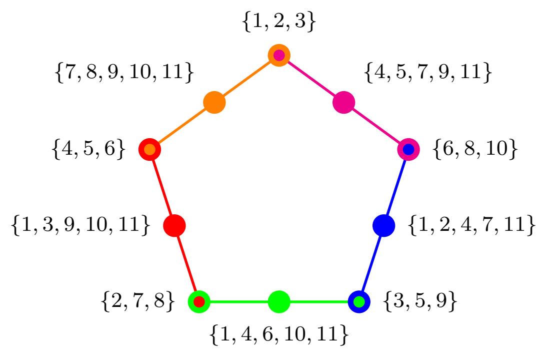

Figure 2: Pentagon survey logic. Each color corresponds to one survey context. Adjacent contexts share exactly one atom, producing the cyclic pentagon structure. The construction is an idealized experimental design for strong framing effects, not a claim that ordinary multi-issue surveys are impossible.

where A_j denotes the jth intertwining atom. Taking the expectation over any probability distribution on S_{11} yields

$$\sum_{j=1}^{5} P(A_j) \leq 2. \tag{6}$$

This is the KCBS-type noncontextual bound for a five-cycle exclusivity structure Bub and Stairs (2009); Klyachko, Can, Binicioğlu, and Shumovsky (2008). In the between-subjects design proposed here, each $P(A_j)$ is estimated as the marginal frequency with which respondents in the appropriate arm fall into the cluster A_j, so that no respondent contributes to two adjacent A_j's simultaneously. No matter how the population is distributed across the eleven profiles, the total intertwining-atom weight cannot exceed 2.

If, however, the respondents' cognitive states are modeled by vectors in a Hilbert space, as in quantum-cognition models Busemeyer and Bruza (2012), then probabilities of the corresponding five cyclic events can obey Born's rule. Under a suitable faithful orthogonal representation of the five-cycle exclusivity structure, one obtains the Lovasz-theta value

$$\sum_{j=1}^{5} P(A_j) \leq \sqrt{5} \approx 2.236. \tag{7}$$

This is about a twelve percent increase over the classical bound. A survey experiment that implements the pentagon exclusivity structure and finds aggregate intertwining-atom probabilities exceeding 2 would falsify the classical latent-partition model and motivate a nonclassical, quantum-like cognitive model. Conversely, satisfaction of the classical bound would support the more parsimonious value-definite interpretation.

Caveats for the empirical test. Because the design is between-subjects, each intertwining atom A_j is in fact estimated from two distinct experimental arms (the two adjacent contexts of which it is an atom). For the KCBS bound to apply as written, an auxiliary *marginal-consistency* or *no-disturbance* condition must hold: the estimated frequency of A_j must agree, within sampling error, between the two arms. A failure of this condition—which is itself an empirically checkable hypothesis—is a generic alternative explanation for any apparent violation of (6), distinct from quantum cognition. Two further alternative explanations must also be controlled for: (i) *selection effects*, including framing-induced differential non-response and panel attrition, which can mimic structure-violating frequencies without any cognitive non-classicality; and (ii) *coarsening leakage*, in which the three-cluster classification of one frame implicitly absorbs information about other issues through correlated item wording, breaking the assumed independence of contexts. A genuine empirical case for quantum-cognitive structure thus requires ruling out, in addition to (6), these classical pathways to apparent bound violation.

5 Scenario 3: Diagnostic complementarity in clinical psychology (Triangle logic)

Our third scenario turns to clinical diagnostics. It shows how cyclic interference between different psychological and physiological assessment tools can instantiate a triangle partition logic. As in Scenario 1, the forced-choice restriction is a deliberate idealization: real multimethod assessment batteries do administer several instruments to the same patient, and the literature on disturbance effects between them is itself contested. Our purpose is to identify the abstract structure that *would* arise if the cyclic disturbance pattern below held to a sufficient degree.

5.1 What are the latent states?

A clinical psychologist suspects that patients presenting with generalized anxiety disorder fall into four latent diagnostic subtypes differing along two dimensions: cognitive worry and physiological arousal. These subtypes are:

- **Latent state** 1 **("Cognitive-somatic"):** High cognitive worry and high physiological arousal.
- **Latent state** 2 **("Cognitive-dominant"):** High cognitive worry but low physiological arousal.
- **Latent state** 3 **("Somatic-dominant"):** Low cognitive worry but high physiological arousal.
- **Latent state** 4 **("Subclinical"):** Moderate levels of both worry and arousal that fall below salient clinical thresholds on either dimension taken alone.

The set of latent states is $S = \{1, 2, 3, 4\}$. Each patient is exactly one subtype; the diagnostic challenge is that no single instrument can identify all four.

5.2 What are the three contexts?

Three diagnostic instruments are available. Each resolves two of the three clear subtypes $1, 2, 3$ as singleton observed categories while conflating the remaining clear subtype with the elusive subtype 4.

- **Context $\mathcal{C}_A$ — Self-report questionnaire.** The patient completes a standardized inventory of worry-related cognitions. Instrument A picks up the cognitive dimension:

$$\mathcal{C}_A = \{\{1\}, \{2\}, \{3, 4\}\}.$$

- **Context $\mathcal{C}_B$ — Behavioral observation.** A clinician observes the patient during a structured social interaction and rates visible signs such as gaze aversion, fidgeting, tremor, and speech latency. Instrument B picks up the behavioral expression of anxiety:

$$\mathcal{C}_B = \{\{2\}, \{3\}, \{1, 4\}\}.$$

- **Context $\mathcal{C}_C$ — Physiological stress test.** The patient undergoes a standardized stress-induction protocol while heart-rate variability and cortisol are recorded. Instrument C picks up the somatic dimension:

$$\mathcal{C}_C = \{\{1\}, \{3\}, \{2, 4\}\}.$$

Table 2: How the four latent diagnostic subtypes map to observed categories under each of the three instruments. Boldface marks singleton observed categories; plain text marks conflated pairs.

Latent state	$\mathcal{C}_A$ (Self-report)	$\mathcal{C}_B$ (Behavioral)	$\mathcal{C}_C$ (Physiological)
1 (Cognitive-somatic)	**severe cognitive**	unremarkable	**severe physiological**
2 (Cognitive-dominant)	**moderate cognitive**	**avoidant**	low physiological
3 (Somatic-dominant)	low cognitive	**tremorous**	**moderate physiological**
4 (Subclinical)	low cognitive	unremarkable	low physiological

Table 2 summarizes the mapping. The crucial pattern is that 4 never forms a singleton. Under every instrument it is conflated with a different partner: with 3 under A, with 1 under B, and with 2 under C. It is the "stealth" subtype that each instrument misses in a different way.

5.3 Why only one instrument per session?

The three instruments interfere with one another in a cyclic pattern that, in the idealized model, prevents sequential administration within a single diagnostic session:

1. *A degrades B.* Completing the self-report questionnaire induces introspective self-focus, which alters the patient's spontaneous behavior in the subsequent social interaction Duval and Wicklund (1972).

2. *B degrades C.* The social interaction of being observed by a rater elevates baseline physiological arousal through social-evaluative threat Dickerson and Kemeny (2004).

3. *C degrades A.* Undergoing a physiological stress-induction protocol triggers mood-congruent recall and somatic focusing, biasing subsequent self-reports Bower (1981).

These interferences are systematic disturbances that change the patient's momentary state in ways relevant to the other instruments. The clinician is then forced to choose one instrument per session. This is a paradigm case of disturbance-based complementarity (C2 in the taxonomy of Section 1): the forced choice arises from the interplay between instruments rather than from any intrinsic limitation of any single instrument.

5.4 What are the intertwining atoms?

Each pair of contexts shares exactly one observed category (Table 3):

- $\{1\}$ is shared by $\mathcal{C}_A$ and $\mathcal{C}_C$. Both the self-report questionnaire and the physiological stress test single out the Cognitive-somatic subtype.

- $\{2\}$ is shared by $\mathcal{C}_A$ and $\mathcal{C}_B$. The Cognitive-dominant subtype is identifiable whether one reads the patient's report or watches the patient's behavior.

- $\{3\}$ is shared by $\mathcal{C}_B$ and $\mathcal{C}_C$. The Somatic-dominant subtype appears through both visible tremor and physiological reactivity.

These three intertwining atoms are the vertices of a triangle in the Greechie diagram (Fig. 3). Each edge of the triangle represents one context; the non-shared observed categories $\{3, 4\}$, $\{1, 4\}$, and $\{2, 4\}$ sit at the midpoints of the edges. The result is Wright's triangle logic (Wright, 1990, Figure 2, p. 900); see also Ref. (Dvurečenskij et al., 1995, Fig. 6, Example 8.2, pp. 414,420,421).

5.5 The exotic probability weight: clinical interpretation

Wright Wright (1978) (see also Ref. (Gerelle et al., 1974, Fig. V)) showed that cyclic logics with an odd number n of intertwining atoms—including the triangle, pentagon, and pentagram logics—support an "exotic" probability weight

Table 3: Intertwining atoms of the triangle logic. Each row names a pair of contexts, the shared observed category, and the clinical reason why the subtype is identifiable under both instruments.

Context pair	Shared atom	Reason for identifiability
$\mathcal{C}_A$–$\mathcal{C}_C$	$\{1\}$	Extreme on both worry and arousal
$\mathcal{C}_A$–$\mathcal{C}_B$	$\{2\}$	High worry shows in report and behavior
$\mathcal{C}_B$–$\mathcal{C}_C$	$\{3\}$	High arousal shows in behavior and physiology

in addition to their standard point-evaluation states. This weight assigns value 1/2 to each cyclic intertwining vertex and lies outside the classical convex hull. It is a probability weight on the abstract pasted logic, not a probability measure induced by any distribution over the intended latent social profiles.

To see this, consider the sum of the weights assigned to the n cyclic intertwining atoms,

$$S(\omega) = \sum_{i=1}^{n} \omega(v_i).$$

Because these atoms are arranged in a cycle where each adjacent pair $\{v_i, v_{i+1}\}$ belongs to a common block, any valid weight must satisfy

$$\omega(v_i) + \omega(v_{i+1}) \leq 1.$$

For the exotic state ω_{ex}, one has

$$S(\omega_{\text{ex}}) = \frac{n}{2}.$$

However, for any dispersion-free $\{0, 1\}$-valued state on the cycle, adjacent intertwining atoms cannot both receive value 1. On an odd cycle, the maximum number of non-adjacent vertices that can receive value 1 is $(n-1)/2$; hence

$$S(\omega_c) \leq \frac{n-1}{2}$$

for every classical deterministic state ω_c. Since every state in the classical convex hull is a mixture of such deterministic states, it also satisfies

$$S(\omega_{\text{hull}}) \leq \frac{n-1}{2}.$$

Because $n/2 > (n-1)/2$, the exotic state lies outside the classical convex hull.

Wright motivated the empirical plausibility of such nonclassical weights by giving a simple consumer-choice example. He wrote (Wright, 1978, p. 272):

> "...suppose that there is a consumer who has no strong opinions one way or the other about the products, so that he is equally likely

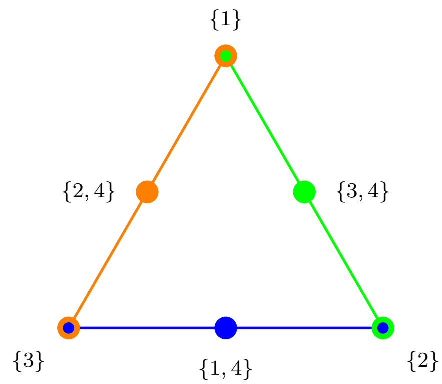

Figure 3: Greechie diagram of the triangle logic applied to clinical diagnosis. Each edge represents one diagnostic context. The corner vertices are the intertwining atoms: the singleton observed categories, each identifiable under two adjacent instruments. The midpoint vertices are the conflated observed categories, each merging the subclinical subtype 4 with a different partner. The six observed categories form a non-Boolean pasted operational logic. They are, of course, representable as subsets of the underlying latent space S; the non-Booleanity concerns the restricted context-wise operations, not the nonexistence of an ambient Boolean algebra.

> to say either yes or no to the first question, but suppose, further, that he is embarrassed about saying no Smith (1975) so that if he says no to the first question, then he always says yes to the second."

The point of Wright's example is not that the consumer has a fixed pre-existing answer to all possible questions. Rather, the response to a question is partly shaped by the local question context and by the transition to the next admissible alternative. The resulting probability assignment can be normalized within each context while failing to arise from a single global distribution over pre-existing answers.

The clinical example introduced in Section 5 provides a concrete interpretation of this exotic triangle weight. Recall that the four latent diagnostic subtypes were denoted by $1, 2, 3, 4$. Subtypes $1, 2, 3$ are the three diagnostically salient or "clear" presentations, while subtype 4 is the subclinical or masked presentation that is never identified as a singleton by any one instrument. A diagnostic context means here one complete diagnostic arrangement—for example, a self-report questionnaire, a behavioral observation protocol, or a physiological stress test. Each such arrangement partitions the same latent patient population into the categories that this particular instrument can distinguish.

In the stylized triangle model, the three diagnostic contexts are

$$\mathcal{C}_A = \{\{1\}, \{2\}, \{3, 4\}\},$$

$$\mathcal{C}_B = \{\{2\}, \{3\}, \{1, 4\}\},$$

and

$$\mathcal{C}_C = \{\{1\}, \{3\}, \{2, 4\}\}.$$

For instance, $\mathcal{C}_A$ may be read as a self-report context: it sharply distinguishes subtypes 1 and 2, but conflates subtype 3 with the masked subtype 4. Similarly, $\mathcal{C}_B$ may be read as a behavioral-observation context: it sharply distinguishes subtypes 2 and 3, but conflates subtype 1 with subtype 4. Finally, $\mathcal{C}_C$ may be read as a physiological-stress context: it sharply distinguishes subtypes 1 and 3, but conflates subtype 2 with subtype 4.

The resulting pasted logic has the form of a triangle. Its three cyclic intertwining atoms are the singleton diagnostic categories

$$\{1\}, \qquad \{2\}, \qquad \{3\}.$$

They are called intertwining atoms because each of them occurs in two different diagnostic contexts: $\{1\}$ occurs in $\mathcal{C}_A$ and $\mathcal{C}_C$, $\{2\}$ occurs in $\mathcal{C}_A$ and $\mathcal{C}_B$, and $\{3\}$ occurs in $\mathcal{C}_B$ and $\mathcal{C}_C$. The remaining atoms $\{3, 4\}$, $\{1, 4\}$, and $\{2, 4\}$ are not shared between contexts. They are context-specific conflations of one clear subtype with the masked subtype 4.

The exotic triangle weight assigns

$$P(\{1\}) = P(\{2\}) = P(\{3\}) = \frac{1}{2}$$

to the three singleton intertwining atoms and assigns probability zero to the three non-shared atoms:

$$P(\{3, 4\}) = P(\{1, 4\}) = P(\{2, 4\}) = 0.$$

This is a valid probability weight on the abstract pasted triangle logic, because each diagnostic context is separately normalized:

$$\mathcal{C}_A : \quad P(\{1\}) + P(\{2\}) + P(\{3, 4\}) = \frac{1}{2} + \frac{1}{2} + 0 = 1,$$

$$\mathcal{C}_B : \quad P(\{2\}) + P(\{3\}) + P(\{1, 4\}) = \frac{1}{2} + \frac{1}{2} + 0 = 1,$$

and

$$\mathcal{C}_C : \quad P(\{1\}) + P(\{3\}) + P(\{2, 4\}) = \frac{1}{2} + \frac{1}{2} + 0 = 1.$$

Thus each instrument, considered by itself, yields an ordinary normalized probability distribution over its three observed categories.

What fails is the existence of a single underlying prevalence distribution over the four intended latent diagnostic subtypes. Suppose the population were described by fixed prevalences

$$(\lambda_1, \lambda_2, \lambda_3, \lambda_4), \qquad \lambda_i \geq 0, \qquad \sum_i \lambda_i = 1.$$

Then the probabilities of the three singleton categories would have to be

$$P(\{1\}) = \lambda_1, \qquad P(\{2\}) = \lambda_2, \qquad P(\{3\}) = \lambda_3.$$

Consequently,

$$P(\{1\}) + P(\{2\}) + P(\{3\}) = \lambda_1 + \lambda_2 + \lambda_3 \leq 1.$$

The exotic triangle weight instead gives

$$P(\{1\}) + P(\{2\}) + P(\{3\}) = \frac{3}{2},$$

which violates this classical latent-prevalence bound. Equivalently, it would require

$$\lambda_1 = \lambda_2 = \lambda_3 = \frac{1}{2},$$

which already sums to $3/2$, before any probability is assigned to subtype 4.

The clinical interpretation must therefore be stated carefully. The exotic weight does not describe an ordinary population in which each patient has one fixed diagnostic subtype drawn from $\{1, 2, 3, 4\}$. Rather, it describes a context-dependent diagnostic response pattern. In each diagnostic arrangement, two singleton categories are available as sharp classifications, and the population is split evenly between them. The remaining clear subtype is not available as a singleton in that arrangement, because it is absorbed into the context-specific conflated atom containing the masked subtype 4. Which pair of singleton classifications is available depends on the context:

$$\mathcal{C}_A \text{ makes } \{1\} \text{ and } \{2\} \text{ sharp},$$

$$\mathcal{C}_B \text{ makes } \{2\} \text{ and } \{3\} \text{ sharp},$$

and

$$\mathcal{C}_C \text{ makes } \{1\} \text{ and } \{3\} \text{ sharp}.$$

The same population can therefore display marginal weight $1/2$ on each of the three singleton diagnoses when these marginals are estimated in the appropriate context arms, even though no single fixed prevalence distribution over the four latent subtypes can reproduce those three marginals simultaneously.

In this sense, Wright's consumer example and the clinical triangle have the same formal moral. The empirical responses are locally well-defined and normalized within each question or diagnostic context, but they need not be marginals of one global distribution over pre-existing answers or pre-existing diagnostic types. If empirical frequencies in a clinical study approximated the exotic triangle weight, the classical latent-partition model would therefore be inadequate.

There are, however, several possible interpretations of such a finding. A quantum-cognition interpretation would say that diagnostic profiles are not merely revealed by the instrument but are partly constituted by the diagnostic context Busemeyer and Bruza (2012). A more conservative interpretation would be that the experimental design has failed to implement the intended triangle logic: the three arms may sample different populations, the instruments may not realize the assumed partitions, or marginal consistency across shared atoms may fail Garg and Mermin (1984). Thus the exotic weight should not be treated as a realistic default model of clinical diagnosis. It is better understood as a sharp benchmark separating ordinary fixed-subtype prevalence models from genuinely context-dependent diagnostic probability assignments.

6 Scenario 4: Espionage and relational encoding—a classical contrast case

Our fourth scenario plays a different role from the previous three. Rather than instantiating yet another non-Boolean partition logic, it provides a deliberate *contrast case*: it isolates what can already be achieved by classical common-cause models with shared preparation and no quantum or otherwise nonclassical resource. The scenario thereby clarifies what is, and is not, distinctive about the previous three structures, and prepares the ground for the maximally complementary auditing example of Section 8. By modeling two spatially separated espionage agents sharing a predefined codebook, we construct a classical analogue of one aspect—and only one aspect—of the Einstein–Podolsky–Rosen setup: separated perfect correlations generated by common preparation.

6.1 What are the types?

An intelligence agency trains agent pairs using a protocol that creates correlations between their cover stories. Each agent pair is prepared in one of four coordination types:

- **Type 1 (“Mirror”):** Both agents memorize clean financial records and clean personal histories. Code: 00.
- **Type 2 (“Split-A”):** Agent A memorizes a clean financial record but a fabricated personal history; Agent B memorizes the reverse. Code: 01.
- **Type 3 (“Split-B”):** The reverse of Split-A. Code: 10.
- **Type 4 (“Contrast”):** Both agents memorize fabricated financial records and fabricated personal histories. Code: 11.

The latent space is again $S_4 = \{00, 01, 10, 11\}$.

6.2 What are the contexts?

Upon capture, each agent faces one of two interrogation modes:

- **Financial background check (F):** The interrogator examines only financial records. The response is the first digit of the code.

- **Personal history interview (H):** The interrogator examines only personal history. The response is the second digit of the code.

An agent subjected to one mode gives answers that are psychologically activated in a way that contaminates a subsequent interrogation in a different mode. Thus only one mode is applied per agent.

For Alice the two interrogation modes define the partitions

$$\begin{aligned} C_F^{(A)} &= \{\{1,2\},\{3,4\}\} \quad \text{(first digit)}, \\ C_H^{(A)} &= \{\{1,3\},\{2,4\}\} \quad \text{(second digit)}. \end{aligned} \tag{8}$$

Bob has the analogous pair of contexts.

6.3 Relational encoding and why it is local

The agency prepares agent pairs with a probability distribution supported only on the two types

$$D = \{01, 10\},$$

as shown in Table 4. The latent space remains S_4, but the preparation distribution is concentrated on this two-element subensemble. Thus D is not a new logic but a selected preparation inside the original four-state classical model.

The relational encoding guarantees that if both agents face the same interrogation mode, their answers are always opposite. When Alice's financial check yields clean, Bob's yields fabricated, and vice versa. This perfect anticorrelation arises entirely from the shared codebook, a classical common cause established before separation.

6.4 What this scenario shows—and what it does not

This example is EPR-like only in the weak sense of separated perfect correlations generated by common preparation. It is *not* Bell-nonlocal and cannot violate a Bell inequality. Its purpose within this paper is twofold. First, it emphasizes that correlation without communication is not uniquely quantum: a classical common-cause distribution over a small latent space already produces separated perfect anticorrelations. Second, by exhibiting the strongest possible classical correlations on this graph, it sets the benchmark above which genuinely

Table 4: Relational encoding of cover stories for agent pairs. Each entry is a two-digit code. The subensemble S selects types with equal digits; the subensemble D selects types with unequal digits.

Sample	Type 1	Type 2	Type 3	Type 4
S (same)	00	–	–	11
D (different)	–	01	10	–

nonclassical phenomena must operate. What is uniquely quantum is not correlation per se but the violation of Bell-type linear constraints across multiple mismatched-context combinations, which the present preparation demonstrably does not produce. Readers familiar with this point may read the scenario as a sanity check: any social-science "quantum-like" claim must be measured against what such classical constructions already deliver.

6.5 Social science significance

This scenario models situations in which two separated actors produce correlated behaviors through shared preparation rather than direct communication:

- **Coordinated deception:** Espionage cover stories.
- **Institutional scripts:** Employees trained in the same corporate protocol independently produce consistent behaviors when facing customers Cyert and March (1963).
- **Shared cultural schemas:** Individuals socialized in the same culture give correlated responses about values and norms because they internalized the same codebook during childhood Bourdieu (1987).

In all cases, the correlations are local and classical. They lie within the classical polytope associated with the common-cause model.

7 Scenario 5: Legal pluralism and overlapping jurisdictions

Our fifth scenario applies partition logics to comparative law. The existence of multiple overlapping legal frameworks creates a formal structure of complementarity: the same action may be classified differently depending on which legal code is applied. This is the clearest example of framing-based complementarity (C3 of Section 1): no disturbance is involved, and the forced choice arises because adjudication takes place within one forum at a time, not because the act of legal classification physically alters the action.

7.1 What are the types?

In a legally pluralistic society—for example, a post-colonial state where statutory law, customary law, and religious law coexist Griffiths (1986); Merry (1988)—a citizen's action can be classified differently depending on the framework. Consider six types of land-use action:

- a_1: Clearing forest for subsistence farming on ancestral land.
- a_2: Clearing forest for commercial logging on ancestral land.
- a_3: Building a house on communally held village land.

- a_4: Building a commercial structure on communally held village land.
- a_5: Grazing livestock on a protected nature reserve.
- a_6: Grazing livestock on privately owned pasture.

Each action is a determinate fact; the question is how it is categorized by different legal codes.

7.2 What are the contexts?

Three legal codes, each classifying actions into three categories, define three contexts:

- **Statutory environmental law ($C_{\mathbf{stat}}$):** Focuses on environmental impact:
$$C_{\text{stat}} = \{\{a_1, a_2\}, \{a_3, a_4\}, \{a_5, a_6\}\}.$$
- **Customary law ($C_{\mathbf{cust}}$):** Focuses on ancestral rights and communal obligations:
$$C_{\text{cust}} = \{\{a_1, a_3\}, \{a_2, a_5\}, \{a_4, a_6\}\}.$$
- **Religious law ($C_{\mathbf{rel}}$):** Focuses on stewardship obligations:
$$C_{\text{rel}} = \{\{a_1, a_2\}, \{a_3, a_5\}, \{a_4, a_6\}\}.$$

7.3 What are the intertwining atoms?

Two atoms are shared:

- $\{a_1, a_2\}$ is an intertwining atom of C_{stat} and C_{rel}: both statutory and religious law treat forest clearing as a single prohibited class.
- $\{a_4, a_6\}$ is an intertwining atom of C_{cust} and C_{rel}: both customary and religious law place these commercial or noncommunal uses in a separate category, although they interpret the category differently.

Table 5: Classification of six land-use actions under three legal codes. Actions in the same cell of a given column are indistinguishable under that code.

Action	Statutory	Customary	Religious
a_1 (subsist. clearing)	Prohibited	Permitted	Prohibited
a_2 (comm. logging)	Prohibited	Restricted	Prohibited
a_3 (house on village)	Restricted	Permitted	Permitted
a_4 (shop on village)	Restricted	Neutral	Discouraged
a_5 (graze on reserve)	Permitted	Restricted	Permitted
a_6 (graze on private)	Permitted	Neutral	Discouraged

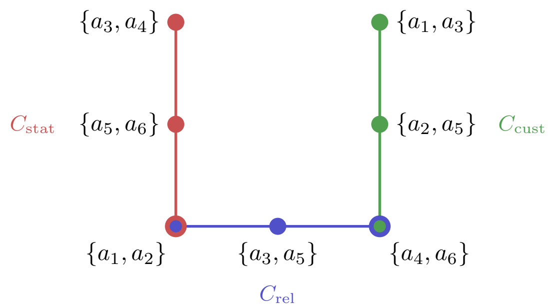

Figure 4: Hypergraph of the legal-pluralism logic. Three legal codes form three contexts. The corner vertices $\{a_1, a_2\}$ and $\{a_4, a_6\}$ are intertwining atoms, each shared by two codes. Other vertices represent groupings specific to one code. The non-Boolean pasting captures the restricted operational structure of forum-dependent classification; the full Boolean refinement over the six action types still exists.

7.4 What does the non-Boolean structure mean for legal practice?

In practice, a land-use dispute is adjudicated under one legal framework at a time: the choice of forum determines which distinctions are drawn and which actions are conflated. Under statutory law, subsistence clearing and commercial logging are treated identically; customary law treats them differently. Under customary law, a house and a shop on village land are different categories; under statutory law both are merely restricted.

The non-Boolean pasted structure does not mean that no common set-theoretic refinement exists. The six action types themselves provide one. Rather, it means that no single legal forum among the three available codes operationally applies all distinctions simultaneously. Each code induces its own coarse graining, and legal outcome depends on the selected forum. In this sense, forum choice is analogous to context choice in a measurement procedure—a pure case of framing-based complementarity without any disturbance dynamics.

8 Scenario 6: Organizational auditing with measurement disturbance

Our final scenario models organizational auditing. It serves two purposes. First, it exhibits *maximal* complementarity at the partition-logic level: the two available audit contexts share no intertwining atom, so the resulting pasted structure is just two disjoint Boolean blocks sharing only $\emptyset$ and the unit. The non-Booleanity is therefore trivial. What carries the substantive content of the example is the second feature: the act of inspection disturbs the corporation's internal state, so that this scenario is the cleanest illustration of disturbance-based complementarity (C2) in our taxonomy. We retain the label "automaton" only in the informal sense that the contexts induce state transitions; a fully formal automaton partition logic in the sense of Schaller and Svozil Schaller and Svozil (1994); Svozil (2005) would specify a transition function $\delta\colon S \times \{a, b\} \to S$ together with output maps, which we sketch but do not develop in detail below.

8.1 What are the types?

A government regulatory agency must assess whether a corporation complies with regulations. The corporation can be in one of four internal organizational states, characterized by two binary dimensions:

- **Financial health:** Sound (0) vs. distressed (1).
- **Operational compliance:** Compliant (0) vs. non-compliant (1).

The four states are:

- $s_1 = (0, 0)$: Financially sound and operationally compliant.
- $s_2 = (0, 1)$: Financially sound but operationally non-compliant.
- $s_3 = (1, 0)$: Financially distressed but operationally compliant.
- $s_4 = (1, 1)$: Financially distressed and operationally non-compliant.

At a fixed time, each corporation is exactly one of these four states.

8.2 What are the contexts?

The auditor can probe the corporation by issuing one of two directives:

- **Financial audit (directive a):** The auditor requests detailed financial records. The response reveals financial health (the first digit), but preparing financial disclosures reallocates organizational resources and may alter operational compliance. The corresponding partition is

$$C_a = \{\{s_1, s_2\}, \{s_3, s_4\}\}.$$

- **Operational inspection (directive b):** The auditor sends inspectors to the factory floor. The response reveals operational compliance (the second digit), but the disruption of the inspection may alter the financial state. The corresponding partition is

$$C_b = \{\{s_1, s_3\}, \{s_2, s_4\}\}.$$

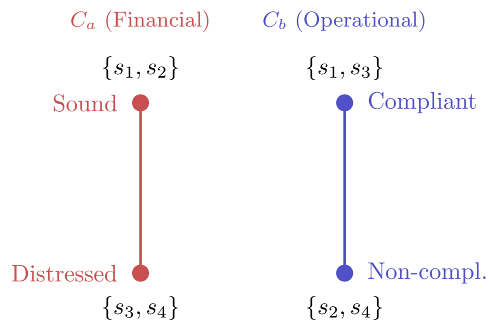

Figure 5: Two complementary audit contexts for the four organizational states. The financial audit partitions the state space by financial health and conflates compliance status. The operational inspection partitions the same state space by compliance status and conflates financial health. The two partitions share no atoms, giving maximal complementarity at the operational level.

8.3 What are the intertwining atoms?

In this scenario, the two partitions share no intertwining atoms:

$$\begin{aligned} C_a &= \{\{s_1, s_2\}, \{s_3, s_4\}\}, \\ C_b &= \{\{s_1, s_3\}, \{s_2, s_4\}\}. \end{aligned} \tag{9}$$

The set $\{s_1, s_2\}$ (sound) is not the same as $\{s_1, s_3\}$ (compliant). Hence no atom appears in both contexts. There is no organizational property in this two-context model that can be ascertained regardless of audit type. This is maximal complementarity in the operational sense—but, as noted above, it is also a structurally trivial case at the level of the pasted logic: with no intertwining atoms, the two Boolean blocks share only $\emptyset$ and S, and the only operational "constraint" is the forced choice between contexts.

8.4 From two-context complementarity to automaton dynamics

What gives this example its substantive content is the *dynamics* induced by the audit directives. We now sketch the additional automaton structure that would be needed to capture this formally.

This formalization connects to Herbert Simon's concept of bounded rationality Simon (1957). An external regulator attempting to understand an organization faces an irreducible limitation imposed by the available probes. At a fixed time, the corporation's state can be represented by an ordinary probability distribution over the four Boolean states $(0,0), (0,1), (1,0), (1,1)$. The nonclassical feature of the auditing model is not the absence of such a distribution, but the fact that probing one coordinate may change the subsequent state of the system.

Formally, the natural extension is a Mealy-style automaton over the state space $S = \{s_1, s_2, s_3, s_4\}$ with input alphabet $\{a, b\}$, output alphabet $\{0, 1\}$, transition function $\delta \colon S \times \{a, b\} \to S$, and output map $\lambda \colon S \times \{a, b\} \to \{0, 1\}$ defined so that $\lambda(s, a)$ returns the first coordinate of s and $\lambda(s, b)$ the second. The disturbance is encoded in δ: a financial audit may flip the operational coordinate ("$\delta(s, a)$ may differ from s in the second digit"), and an operational inspection may flip the financial coordinate. The two-context partition logic above is then the snapshot view at any fixed time; the full empirical content emerges from sequences of directives.

This is the measurement-disturbance problem of quantum mechanics realized here by a completely classical mechanism: the audit process itself consumes resources and changes the organization. A systematic development of the resulting automaton partition logic, including the order effects across sequences of directives and the non-commutativity of a and b at the level of observed outputs, follows the template of Refs. Schaller and Svozil (1994); Svozil (2005) and is left for future work. The point we wish to make here is only that this scenario clearly exhibits disturbance-based complementarity (C2) even though its instantaneous partition logic (C1) is trivial.

9 Discussion: Probability models and quantum cognition

The preceding examples show how partition logics can be instantiated across several social-science domains. We now return to the probability question. The central lesson is that operational complementarity, logical non-Booleanity, and nonclassical probability are distinct notions.

9.1 Summary of the layered interpretation

All six scenarios share a common formal skeleton: a finite latent state space S, together with several partitions of S representing available observational contexts. Three levels must be kept separate:

1. **The latent Boolean space.** At the ontological level of the model, all latent states are ordinary elements of S, and all subsets belong to the Boolean algebra 2^S. In this sense the model is fully value-definite.

2. **The operational pasted logic.** The observer does not have operational access to all distinctions in 2^S. Only the atoms belonging to available contexts are observed, and only context-wise Boolean operations are operationally licensed. The resulting pasted event structure can be non-Boolean even though it is represented by subsets of S.

3. **Probability models.** Classical probabilities arise from distributions over S and hence from convex mixtures of point-evaluation states. Abstract two-valued weights, Born-rule probabilities, and exotic fractional weights

may also be studied on the same or related pasted structures, but they need not correspond to distributions over the intended latent social profiles.

Crosscutting these three levels are the three notions of complementarity (C1)–(C3) introduced in Section 1. Logical non-Booleanity (C1) belongs to level 2; disturbance-based complementarity (C2) is a dynamical statement that goes beyond a fixed-time partition logic and lives most naturally at the automaton/transition level (Section 8.4); framing-based complementarity (C3) typically belongs to level 1, in that different contexts impose different coarse-grainings of an unchanged latent space. The six scenarios are therefore not interchangeable instantiations of a single phenomenon: they illustrate different combinations of (C1)–(C3) on top of the same formal skeleton.

9.2 Why this matters for social science

The resource determines the probability theory. The same exclusivity graph can be realized by a classical urn, a finite automaton, a social-science black box, or a quantum system. A Wright urn or a determinate latent-profile model yields classical probabilities. A quantum system can yield probabilities in a Lovasz-theta-type set. An abstract odd-cycle logic may also support exotic fractional weights. The graph alone does not settle the matter; the realizing resource does.

Complementarity does not imply contextuality. All six scenarios feature complementarity: the inability to perform all measurements simultaneously on an unchanged system. Yet the intended social systems remain value-definite. The applicant has a competence profile, the patient has a subtype, the legal action is a determinate action, and the organization has an internal state. These partition models therefore illustrate that complementarity does not imply Kochen–Specker contextuality Kochen and Specker (1967). The relevant partition logics possess sufficiently many point-evaluation states to separate the intended latent profiles. Their non-Booleanity is operational and epistemic, not ontic.

Testable predictions and their alternative explanations. The distinction becomes empirically important when probabilities are measured. Consider a social-measurement experiment implementing a five-cycle exclusivity structure. If the observed probabilities of the five cyclic events satisfy the classical bound (6), then a latent partition model remains viable. If they exceed the classical bound but remain within the quantum-like bound (7), then a quantum-cognition model gains support. If they violate even the quantum-like bound, then either the intended exclusivity structure has not been implemented or a still different probabilistic model is required.

This three-way contrast (classical / quantum-like / beyond) is what gives the framework its empirical content, but it must be guarded against several standard alternative explanations for any apparent violation of the classical bound, none of which require any quantum-cognitive ontology:

1. *Failure of marginal consistency (no-disturbance).* In a between-subjects design, each intertwining atom is estimated twice, once in each of the two adjacent context arms. A discrepancy between the two estimates breaks the inequality derivation of (6) and can produce apparent violations without any nonclassical resource.

2. *Selection effects.* Framing-induced differential non-response, panel attrition, and self-selection into arms can bias marginal frequencies. A pentagon experiment in which respondents in the "defense" arm differ systematically in baseline attitudes from those in the "immigration" arm cannot be analyzed as if all arms sample the same latent population.

3. *Frame leakage and contextual coarsening.* If the wording of items in one frame implicitly references content relevant to another frame, the assumed independence of contexts breaks down. The same applies if the three-cluster classification of responses is itself data-dependent rather than fixed in advance.

4. *Mis-specified exclusivity structure.* The graph G assumed in the analysis must match the operationally realized exclusivity relations. If two atoms claimed to belong to a common context in fact correspond to non-exclusive empirical events, the classical bound does not apply.

A genuine empirical case for quantum-cognitive structure thus requires not only an apparent bound violation but also a positive demonstration that none of (1)–(4) is responsible. Pre-registered designs that include explicit marginal-consistency checks, balanced randomization, and pre-specified clustering rules are essential for this purpose.

Ontological or measurement-induced uncertainty. So far, we have presumed that uncertainty and complementarity arise because underlying properties are latent and observational means are limited. This is a purely epistemic stance. One may instead ask whether some social or cognitive properties are not merely hidden but partly constituted by the act of measurement. Such behavior would require a stronger formalism than classical partition logic, perhaps involving Kochen–Specker-type contextuality Kochen and Specker (1967) or dynamical models in which the act of measurement changes the state. The organizational audit example already points in this direction: its automaton aspect arises not from the absence of latent states but from state transitions induced by probes.

Non-uniform hypergraphs and varying block sizes. As noted in Section 2.2, the six scenarios presented here use uniform hypergraphs: every context yields the same number of mutually exclusive observed categories. In the social sciences, however, there is no a priori reason for such regularity. A job interview might distinguish three outcome categories while a psychometric test distinguishes five. Formally, the blocks of the partition logic may be Boolean algebras of different orders Greechie (1971). This does not undermine the basic framework: the admissibility axioms of additivity and normalization apply block by block. What does require additional work is the analysis of faithful orthogonal representations and the associated probability polytopes for such non-uniform structures. That systematic study is left for future work.

10 Conclusion

We have shown how partition logics—non-Boolean operational structures originally developed in the foundations of quantum mechanics, generalized urn models, and automata theory Svozil (2005)—can be instantiated in five stylized social-science scenarios, with a sixth (Section 6) included as a deliberate classical contrast case. In each non-contrast scenario, a finite set of underlying types is partitioned by multiple incompatible observational procedures: assessment instruments, survey frames, diagnostic tests, legal codes, or audit directives. The resulting structures exhibit complementarity in one or more of the three senses (C1)–(C3) distinguished in Section 1: no single procedure reveals all relevant distinctions. Yet every system has a fully determinate type. The limitation is epistemic and operational, not ontic.

The social-science scenarios therefore occupy a position between ordinary Boolean description and genuinely quantum contextuality. They have non-Boolean operational logics and complementarity, but retain full value definiteness Svozil (2020). This demonstrates concretely that complementarity does not imply Kochen–Specker contextuality. A pasted operational event structure may fail to be Boolean under its restricted operations even though it is represented inside an ambient Boolean algebra of latent states.

At the same time, the mathematical fact that many such structures also admit quantum-like or exotic probability weights opens a door to empirical testing. If human cognitive or social systems produce probability distributions that violate the classical convex-hull bounds but respect quantum-like or other admissible bounds—and the classical alternative explanations enumerated in Section 9 have been ruled out—the partition-logic framework provides the language in which to formulate and test this hypothesis. Conversely, if the classical bounds are respected, the partition logic with its convex hull of point-evaluation states provides a complete and parsimonious account of the observed complementarity—no quantum formalism is required.

The development of experiments that probe these questions—for instance, large-scale pre-registered surveys with carefully designed pentagon structures, including explicit marginal-consistency checks, or clinical assessment batteries with triangle-logic configurations—is a promising direction for future work at the intersection of formal logic, quantum foundations, and social science methodology.

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